Global Thresholding and Morphological Processing of Rice Grain Images

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Name: Farrel Anggito Baswara Marpaung
NIM: 25/559096/PA/23506

1 Theoretical Background

1.1 Global thresholding

For grayscale intensity I(r, c), a pixel is assigned to the foreground when its value is at least T. The same T is used throughout the image. In this experiment, T = 128 and intensity values range from 0 to 255.

$$B(r,c) = \begin{cases} 1, & I(r,c) \geq 128, \\ 0, & I(r,c) < 128. \end{cases}$$

White represents foreground (1), and black represents background (0). This polarity is suitable for bright grains on a darker surface. A manually selected threshold is distinct from an automatically estimated threshold or a spatially adaptive threshold.

1.2 Structuring element

The origin of the 3 × 3 cross is its centre. The selected offsets are the centre and its four horizontal and vertical neighbours. Ones select positions; zeros mean those positions are ignored. A zero is not a requirement that the corresponding image pixel be black.

```
se = [
    [0, 1, 0],
    [1, 1, 1],
    [0, 1, 0]
]
```

The lecture illustrates square masks and also describes other shapes. A cross is therefore a valid choice. Compared with a square of the same bounding size, it checks fewer positions: erosion is less restrictive, and dilation adds fewer diagonal neighbours. The relative effect of composite opening or closing still depends on the object geometry.

1.3 Basic operations

Operation	Decision or sequence	Typical effect
Dilation	At least one selected neighbour is 1	Expand foreground regions
Erosion	Every selected neighbour is 1	Shrink foreground regions
Opening	Erosion, then dilation with the same SE	Remove small foreground features
Closing	Dilation, then erosion with the same SE	Fill small background features

Opening and closing follow the lecture definitions. Their effects depend on whether a feature can accommodate the chosen SE. Closing does not fill every hole, and opening does not preserve every true object. The square and cross masks discussed here are symmetric, so mask reflection does not change them.

2 Experimental Method

2.1 Input and environment

The input is the rice-grain sample selected from the MathWorks illumination-correction example. Google Colab is used to upload the image and execute Python code. Pillow opens the image and converts it to grayscale, NumPy converts pixel data to a list of rows, and Matplotlib displays the outputs. Thresholding and morphology use manual Python functions rather than library segmentation routines.

Setting	Value
Threshold	128 with foreground defined by intensity $\geq$ threshold
Foreground and background	1 and 0, respectively
Structuring element	3×3 cross with its origin at the centre
Boundary treatment	Positions outside the image are treated as 0
Comparison design	Each operation starts from the same binary image
Evaluation	Qualitative inspection of saved notebook figures

2.2 Processing procedure

1. Upload one image and display the original.
2. Convert the image to 8-bit grayscale and apply global thresholding.
3. Display and retain the thresholded image as the baseline.
4. Apply dilation, erosion, opening, and closing independently to that baseline.
5. Display each result separately and inspect specks, grain boundaries, gaps, and missing fragments.

2.3 Implementation details

The helper get_neighbours aligns the SE centre with the current pixel. It collects only positions whose mask value is 1. Out-of-bounds positions contribute 0. Dilation uses int(any(neighbours)); erosion uses int(all(neighbours)). Each function writes into a new image so changes made during a pass do not influence later decisions in the same pass.

Opening and closing call the two basic functions in opposite orders. In the comparison, closed_image is computed from binary_image, not from opened_image. The experiment therefore isolates each operation rather than presenting four successive filters.

For an $H \times W$ image and a mask containing K positions, each basic pass takes $O(HWK)$ time. With a fixed 3×3 mask this is $O(HW)$. Opening and closing each use two passes. Output storage is $O(HW)$, with a small neighbour list for each evaluated pixel.

3 Original Image and Thresholding Result

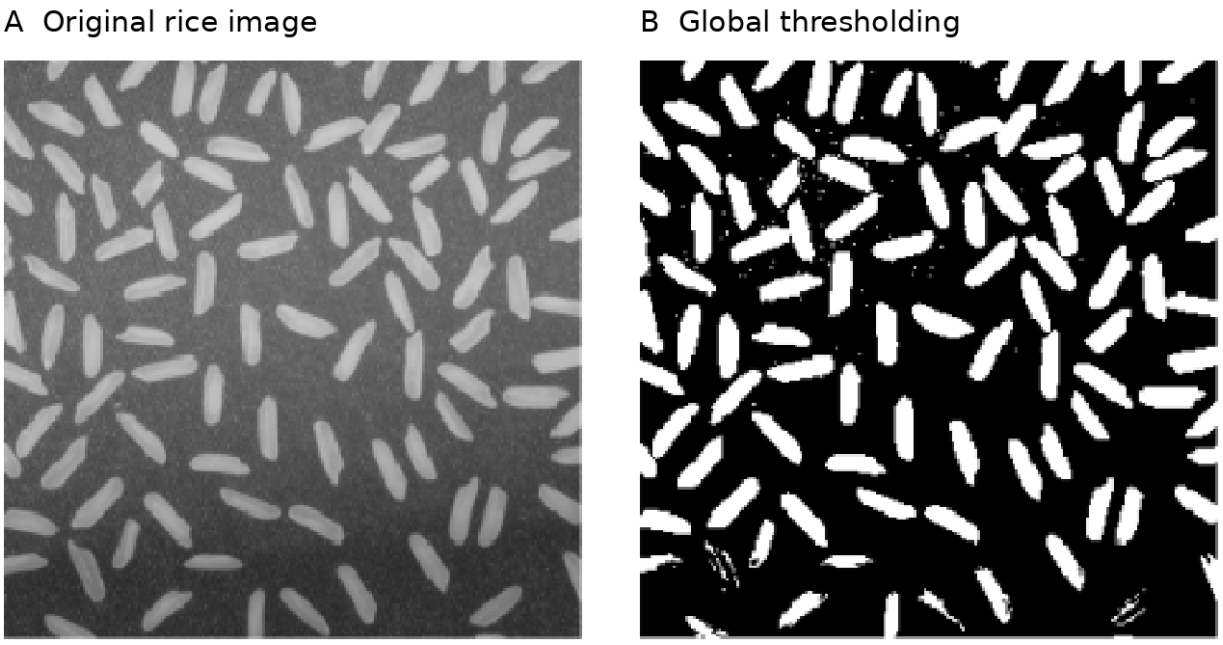


Figure 1: Original rice image and global thresholding baseline at T = 128. Rice regions are cropped from the saved notebook figures for illustration; this cropping is not an additional preprocessing step in the experiment.

3.1 Original image

The original image contains elongated bright grains against a grey background. The lower part is visibly darker than the upper region. Several grains touch the image boundary. These conditions matter because the threshold depends on intensity, while later morphological decisions depend only on the resulting binary shape.

3.2 Thresholded baseline

Most clearly illuminated grains become white, and much of the darker background becomes black. Small white specks and jagged or fragmented features remain. Near the bottom, some grains visible in the original appear only as partial white traces or are largely absent. The darker portions have fallen below the chosen threshold.

These missing regions should be distinguished from noise removal. A morphology result may look cleaner after deleting fragments while becoming less complete as a representation of all grains. Consequently, visual smoothness alone is not sufficient evidence of improved segmentation.

3.3 Evaluation scope

The results below are observations from the notebook's saved display figures. The original binary arrays and a manually labelled reference mask were not exported with these figures. No pixel-accuracy, overlap score, grain-count accuracy, or execution-time value is therefore reported. Display margins and image resampling also make screenshot-based pixel counts unsuitable as exact algorithm measurements.

4 Morphological Results

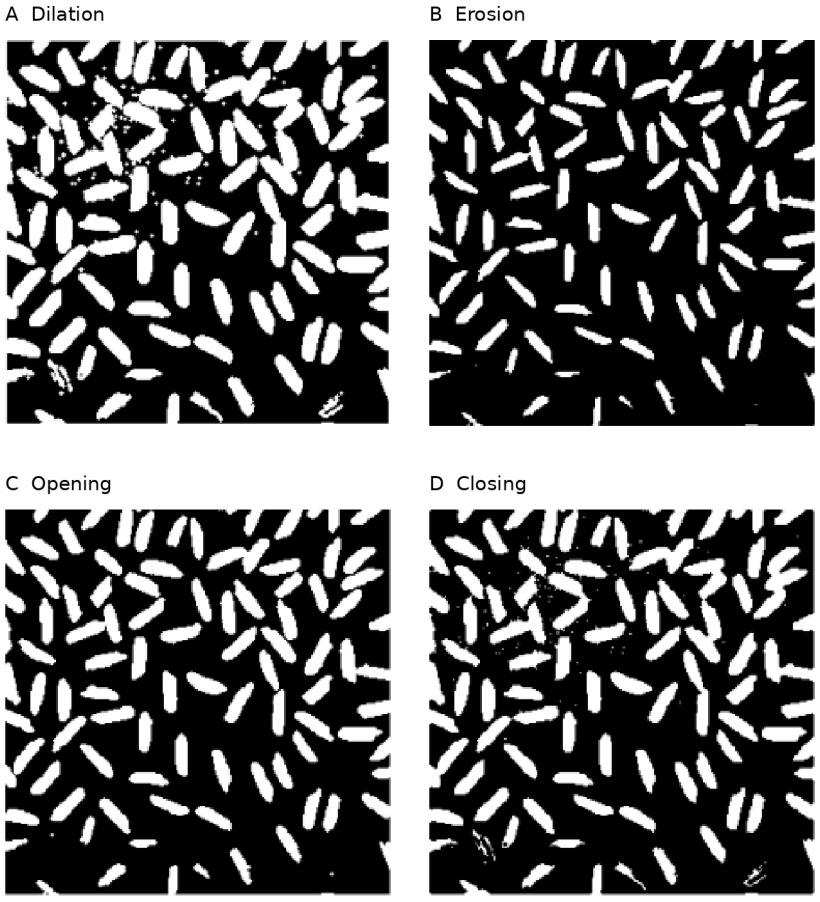


Figure 2: Independent results using the 3×3 cross SE. Each panel starts from the thresholded baseline. Rice regions are cropped from the saved figures, without applying further morphology.

Result	Visible observation	Interpretation
Dilation	Grains and small fragments appear thicker; some gaps narrow.	Foreground grows, including unwanted fragments.
Erosion	Grains become thinner; many small lower-edge traces disappear.	Selected neighbours must all be white.
Opening	Small specks and fine fragments are reduced; larger grains remain recognisable.	Small foreground features do not survive erosion and regrowth.
Closing	Some small black interruptions narrow or fill; nearby white features become more connected.	Background features smaller than the mask can be suppressed.

The images demonstrate different shape changes rather than four levels of progressively better segmentation. Small gaps and boundary effects should be inspected at the original output scale when choosing an operation for measurement or counting.

5 Conclusion

The implementation successfully demonstrates global thresholding followed by independent dilation, erosion, opening, and closing. The functions follow the lecture definitions for a centred symmetric structuring element. The separate displays make it possible to identify the effect of each operation without confusing it with the effect of a preceding filter.

The rice example shows that dilation expands foreground, erosion removes thin and small features, opening suppresses some small white fragments, and closing fills some narrow black gaps. Opening is the most relevant of these operations when the immediate purpose is reducing small white foreground features, but the figures alone do not establish that it is the most accurate segmentation. Missing dim grains and possible joining of nearby grains remain important limitations.